UNIVERSAL FIRST ORDER THICKENINGS COMMUTE WITH ETALE LOCALIZATION ON SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3308-3325 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3327-3334, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete).

Editorial note. BORDERLINE (11th): proof body is three exact in-source citations with the conclusions restated — kept under the written-reduction reading pending your BORDERLINE verdict; it is the localization input used by schemes-side card L16216's conormal argument.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas formally-unramified, universal-thickening-over-formally-etale, etale-morphism-of-universal-thickenings.

Source reference. Original lemma 0605: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3025–3040. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04G8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2724–2742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0604: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3004–3010. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Taking the universal first order thickenings commutes with étale localization. More precisely, let $h : Z \to X$ be a formally unramified morphism of*

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algebraic spaces over a base scheme S. Let

$$\begin{array}{ccc} V & \longrightarrow & U \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

be a commutative diagram with étale vertical arrows. Let Z' be the universal first order thickening of Z over X. Then $V \to U$ is formally unramified and the universal first order thickening V' of V over U is étale over Z'. In particular, $\mathcal{C}_{Z/X}|_V = \mathcal{C}_{V/U}$.

Proof. The first statement is Lemma 04G8. The compatibility of universal first order thickenings is a consequence of Lemmas 0604 and 0605. □